

NEBULAOPS v22.3 FINAL

Backend & API Technical Specification

Backend service catalog, gateway contracts, request/response examples and API closure status.

18

backend modules

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OpenAPI files

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new API services

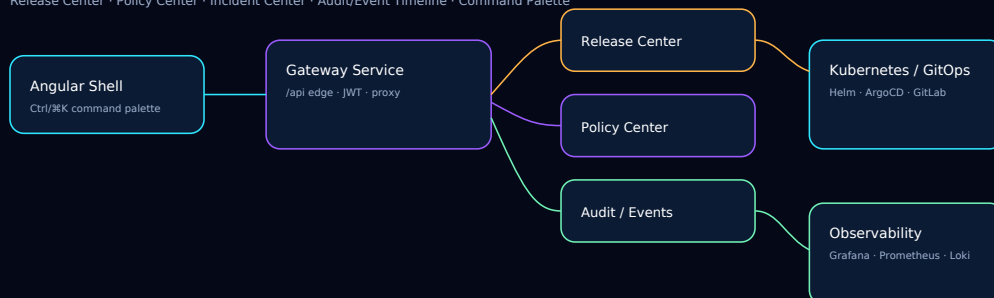
1

gateway edge

Professional release closure package - prepared for technical review, handover and local validation.

NebulaOps v22.3 · Operational Intelligence & Governance Layer

Release Center · Policy Center · Incident Center · Audit/Event Timeline · Command Palette



Reference figure - NebulaOps v22.3 platform topology.

1. Backend/API closure

The backend closure consolidates the v22.3 services into a coherent API surface through gateway-service. Existing backend services are preserved and the new release/policy/audit/notification flows are exposed under /api routes for shell and micro frontend consumption.

Module	Role	Key endpoints / notes
ai-ops-service	Domain/runtime service	See service controllers and OpenAPI stubs where available
audit-service	Audit trail and platform event timeline	See service controllers and OpenAPI stubs where available
auth-service	Domain/runtime service	See service controllers and OpenAPI stubs where available
cost-analytics-service	Domain/runtime service	See service controllers and OpenAPI stubs where available
devsecops-service	Domain/runtime service	See service controllers and OpenAPI stubs where available
environment-manager-service	Domain/runtime service	See service controllers and OpenAPI stubs where available
file-service	Domain/runtime service	See service controllers and OpenAPI stubs where available
gateway-service	/api edge, proxy integration, runtime and docs endpoints	See service controllers and OpenAPI stubs where available
gitops-control-service	Domain/runtime service	See service controllers and OpenAPI stubs where available
nebulaops-shared-kernel	Shared contracts, responses, events, security constants	See service controllers and OpenAPI stubs where available
notification-service	Notifications, preferences, mark-read and SSE stream	See service controllers and OpenAPI stubs where available
observability-service	Domain/runtime service	See service controllers and OpenAPI stubs where available
pipeline-engine-service	Domain/runtime service	See service controllers and OpenAPI stubs where available
policy-governance-service	Policy catalog and PASS/WARN/FAIL evaluation	See service controllers and OpenAPI stubs where available
release-orchestrator-service	Release lifecycle, promote/rollback, timeline, health impact	See service controllers and OpenAPI stubs where available
spring-mvc-service	Domain/runtime service	See service controllers and OpenAPI stubs where available
task-service	Domain/runtime service	See service controllers and OpenAPI stubs where available
terraform-studio-service	Domain/runtime service	See service controllers and OpenAPI stubs where available

2. New API contracts

Domain	Method/path	Purpose
Release	GET /api/releases	List release candidates and rollout state
Release	POST /api/releases/{id}/promote	Accept release promotion request
Release	POST /api/releases/{id}/rollback	Queue rollback action
Policy	GET /api/policies	List governance policies
Policy	POST /api/policies/evaluate	Evaluate deployment/security/cost gates
Audit	GET /api/events	Read platform event timeline
Audit	POST /api/events	Record platform event
Notifications	GET /api/notifications/live	Read live notification inbox
Notifications	GET /api/notifications/stream	SSE notification stream
AI Ops	POST /api/ai-ops/incidents/analyze	Analyze incident and produce RCA/runbook input
DevSecOps	POST /api/devsecops/scan/image	Image scan contract
Cost	GET /api/cost/forecast	FinOps forecast contract
Kubernetes	POST /api/kubernetes/resources/diff	Diff resource before apply

3. Request/response examples

POST /api/policies/evaluate
Content-Type: application/json

```
{
  "target": "release rel-22.3-001",
  "environment": "local",
  "image": "nebulaops-v22-3-gateway-service:latest"
}
```

Response 200

```
{
  "id": "eval-...",
  "target": "release rel-22.3-001",
  "status": "WARN",
  "results": [
    {"policyId": "pol-no-latest", "status": "PASS"},
    {"policyId": "pol-critical-vuln", "status": "WARN"}
  ],
  "evaluatedAt": "..."
}
```

POST /api/ai-ops/incidents/analyze
Content-Type: application/json

```
{
  "source": "kubernetes",
  "affectedService": "gateway-service",
  "symptoms": ["HTTP 502", "pod restart", "latency spike"]
}
```

Response 200

```
{
  "severity": "HIGH",
  "rootCause": "Downstream service degradation detected",
  "evidence": ["Loki query", "Prometheus metric", "Kubernetes event"],
  "recommendations": ["Check downstream health", "Inspect recent release"]
}
```

4. OpenAPI assets

OpenAPI file	Scope
audit-service.openapi.yml	v22.3 API documentation stub
policy-governance-service.openapi.yml	v22.3 API documentation stub
release-orchestrator-service.openapi.yml	v22.3 API documentation stub